

Poster: GEM-PLI Side Channel in GPON: Inferring Encrypted Streaming Traffic

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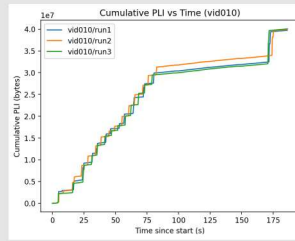
Title: Protocol vulnerability in Gigabit Passive Optical Networks (GPON)

This poster presents a preliminary analysis showing that the Payload Length Indicator (PLI), which remains in plaintext in GEM headers, may be exploited as a side channel that enables inference of streaming activity. Using a GPON downstream simulator, We demonstrate that PLI traces are highly consistent across repeated runs of the same video and suggest that a passive attacker on the same optical splitter as the victim could potentially infer a victim's streaming activity (e.g., Youtube Video Identification)

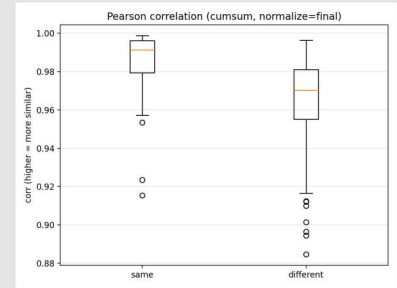
INTRODUCTION

As FTTH(Fiber-To-The-Home) Internet has become popular, it is important to ensure traffic confidentiality. GPON(Gigabit Passive Optical Networks) supports AES based downstream traffic encryption but some fields in GEM(GPON Encapsulation Method) frame remain Plaintext. You may think such fields is not important. However recent works demonstrated that such fields can be exploited to inter user's video streaming activity(e.g., Leaking the name of the video user is watching) In this poster, We Present preliminary evidence that PLI values in GEM header can be exploited to inter a victim's video activity.

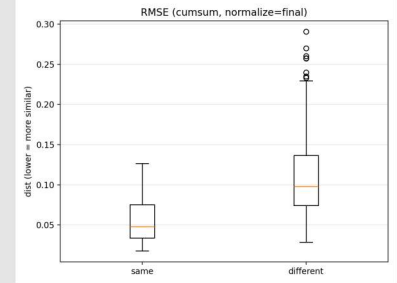
EXPERIMENT RESULTS



Cumulative PLI - Time Graph for one video showing that greatly identical slopes between three runs, PLI - Time Graph

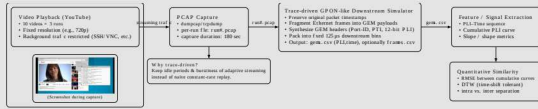


Pearson correlation (higher=more similar)
AUC=0.779



RMSE (lower=more similar)
AUC=0.826

EXPERIMENT



1. we developed a software GPON downstream simulator that reproduces the process of receiving GTC frames, extracting GEM frames, and exporting plaintext GEM header fields into datasets.
2. At first we played each of 10 videos three times(RUN1, RUN2, RUN3) in our linux based system to produce packet capture datas (PCAPs) where other streaming traffic(e.g., VNC) is restricted to minimize interference.
3. Then we simulated that PCAPs packet capture file to records the resulting PLI values and timestamps into CSV traces dataset.
4. After producing dataset we visualized cumulative PLI - Time Graph(Please Look at right side)
5. Also we simulated similarity on the normalized cumulative PLI time series(Using nearest matching over Pearson and RMSE)

TABLE I
PER-QUERY TOP-1 PREDICTIONS. [3]

Query	Prediction	CORRECT	margin	r / RMSE
vid001/r3	vid001/r1	T	2	0.998/0.025
vid002/r3	vid002/r1	T	5	0.995/0.041
vid003/r3	vid003/r2	T	1	0.980/0.050
vid004/r3	vid004/r1	T	2	0.996/0.037
vid005/r3	vid005/r2	T	2	0.998/0.020
vid006/r3	vid008/r1	F	2	0.994/0.048
vid007/r3	vid007/r1	T	2	0.999/0.025
vid008/r3	vid008/r1	T	2	0.999/0.018
vid009/r3	vid009/r2	T	2	0.990/0.041
vid010/r3	vid010/r1	T	3	0.998/0.020

In rank-vote fusion of Perason and RMSE, We Achieve 90% Top-1 Video Identification (*Query: Answer, Prediction: Predicted value obtained through RMSE and Pearson correlation, CORRECT: Is that Correct Prediction(True/False)

r1=RUN1

r2=RUN2

r3=RUN3

We used RUN3 as query and RUN1,2 as database.

These results suggest that encrypting payload bytes alone is insufficient for privacy in GPON, motivating mitigations such as traffic shaping and metadata obfuscation.

CONTRIBUTIONS

- access-layer side channel in encrypted GPON: We identify and empirically study the GEM-header PLI field (plaintext) as a threat source that can enable streaming inference even when GEM payloads are AES CTR encrypted.
- E2E attacker observation pipeline: We formalize a same-splitter adversary who (i) passively captures downstream traffic, (ii) parses GTC/GEM frames, (iii) filters by the victim's Port-ID, and (iv) obtains timestamped PLI traces for database matching-based video identification.
- Reproducible trace generation via a GPON downstream simulator: We build a software trace generator that follows key ITU-T G.984.3 TC-layer parameters relevant to PLI (e.g., 125 μ s framing, GEM header formatting, and fragmentation under the 12-bit PLI limit) to synthesize attacker-visible PLI sequences from encrypted downlink PCAPs.
- Evidence of temporal consistency and distinctiveness at the GEM layer: Across repeated runs of the same YouTube video, cumulative PLI over time traces exhibit consistent slopes, while different videos tend to be separable under similarity metrics (Pearson correlation / RMSE).
- Preliminary video identification results: Using nearest-neighbor database matching with rank voting over Pearson and RMSE, we achieve 90% Top-1 accuracy (9/10) when matching run3 queries against run1,run2 traces on 10 videos.

FUTURE WORK & LIMITATION

- Simulator-driven results $\rightarrow$ hardware validation needed. Our current evidence is obtained from a software GPON downstream trace generator driven by encrypted downlink PCAP replays. As next steps, we will validate the attack end to end on a real GPON testbed (Real OLT/ONU and an optical splitter) using an FPGA based optical capture setup to parse downstream GTC and GEM frames.
- Port-ID assumption and field feasibility. We assume the attacker can obtain and reliably track the victim's Port-ID to filter GEM frames. Future work will study practical Port-ID robustness in the field (e.g., stability across time, vendors, and operational conditions).
- Scale and diversity of evaluation. The current dataset is limited (10 videos, three runs each, fixed 720p, 180s captures). We will expand to more videos and longer sessions, and systematically vary resolution, bitrate, ABR behavior to quantify how identification accuracy degrades or improves under realistic streaming dynamics.
- Real splitter confounders not fully captured. A real PON may include cross-subscriber multiplexing noise, OMCI, and vendor's scheduling/packing artifacts. Future measurements on a shared splitter will characterize these confounders and test mitigation sensitivity.

REFERENCES

- [1] ITU-T Recommendation G.984.3, Gigabit-capable passive optical networks (GPON): Transmission convergence layer specification.
- [2] Cisco, "Understand GPON Technology," Troubleshooting TechNotes. (Document ID: 216230)
- [3] Our Github Repository - <https://gitshare.me/repo/c8afd178-a072-4cee-a01c-8b3d12467a28>
- [4] S. Bae, et al., "Watching the Watchers: Practical Video Identification Attack in LTE Networks," in 31st USENIX Security Symposium (USENIX Security '22), 2022.
- [5] X. Zhang, et al., "Traffic spills the beans: A robust video identification attack against YouTube," Computers & Security, vol. 137, 2024, Art. 103623.
- [6] N. Hu et al., "Breaking Through the Diversity: Encrypted Video Identification Attack Based on QUIC Features," in ESORICS 2024, 2024.



Our Simulator's Code and dataset are publicly available in our Github repository.